

PRIMARY HEALTH COLLECTION

Benchmark Report

Final evaluation of transcription, controlled-noise robustness and local-only inference.

Sauti Yetu Project | Swahili-English healthcare speech | Kenya

Collection origin. This benchmark uses a primary dataset collected by the Sauti Yetu team. The collection was designed to represent code-switched Swahili-English communication in healthcare, where speakers may move naturally between Swahili and English medical terms. It combines 20 health-complaint recordings and 14 WhatsApp voice notes with 30 short triage phrases, including both mixed-language and monolingual controls. The findings describe this team-collected benchmark and are not a population-wide estimate.

Final noise sweep: complete

The benchmark contains **768 of 768 valid noise results**: 64 clips, four SNR levels and three models. There are **zero recorded noise-run errors**.

Principal findings

Transcription. Sahara has the lowest mean word error rate (WER) on the latest unperturbed baseline: **7.44%**, compared with **37.91%** for MMS and **95.89%** for Whisper. This is a comparison of the recorded configurations on this collection, not a general ranking of the model families. [N]

Noise robustness. Sahara remains lowest at every tested noise level. At 5 dB SNR, mean WER is **19.45%** for Sahara, **49.84%** for MMS and **97.92%** for Whisper. All means now use the same 64-clip denominator. [N]

Local-only inference. Both local model paths completed all 64 clips. MMS has lower mean WER and 50 outright clip wins; Whisper has 7 outright wins, with 7 ties. These are local-only execution results, not proof of network-isolated operation. [C]

What is finalized

This report finalizes the available benchmark evidence, with complete noise coverage and clearly separated run modes. It preserves the measured scores and retains the remaining reference, annotation and reproducibility limitations. It does not certify deployment readiness.

The benchmark combines the complete noise run with the local-only evaluation. Run-specific details and calculation rules are summarized in the appendix.

1. Evaluation scope and methods

The evaluation uses the same 64 unique filenames and matching metadata in both retained source exports. The collection contains approximately 12.1 minutes of audio, with clips from 2.04 to 30.96 seconds. References are identified in the manifests as human ground-truth transcripts; this review did not independently listen to the audio. [N, C]

Models: Intron Sahara (remote API), Meta MMS (local; mms-1b-all) and OpenAI Whisper (local; checkpoint not specified in the exports).

Evidence layer	Source and execution flags	Valid results	Use in this report
Latest baseline	13:24:40 UTC --noise --tag noise	64 x 3 = 192	Unperturbed ASR and same-run noise comparison.
Controlled noise	[N] noise_robustness	64 x 4 x 3 = 768	Complete 30, 20, 10 and 5 dB results.
Core local pass	08:11:12 UTC --offline --tag core	64 x 3 = 192	Clean comparison and local-only inference.
Local-only pass	offline_simulation	64 x 2 = 128	Whisper and MMS local inference. Sahara is not included.

Collection composition

Recording group	Clips	File / device metadata	Interpretation
Short triage phrases	30	MP3; laptop microphone; studio quiet	Includes monolingual and mixed-language controls.
Health complaints	20	M4A; smartphone; indoor quiet	Recorded healthcare narratives.
WhatsApp voice notes	14	OGG; smartphone PTT; ambient smartphone note	Channel and speech-style variation.

Recording groups follow the supplied metadata. In particular, the label synthetic_triage_phrase does not independently establish how the audio was generated. All clips are tagged Swahili-English, but that tag alone does not imply that every utterance mixes languages.

Metric and denominator rules

Metric	Interpretation
WER / CER	Word / character edit error relative to the stored reference. Reported means are unweighted clip averages, not pooled corpus error. Lower is better; insertions can make WER exceed 100%.
Latency / RTF	Recorded elapsed seconds and elapsed time divided by audio duration. RTF below 1 means faster than audio duration for that result, not necessarily interactive response speed.
Switch-point WER	Stored error in annotated language-boundary windows: 32 windows across 32 clips. It is not a whole-language accuracy score.

2. Latest baseline transcription

The baseline results in the final noise export are the primary ASR values in this edition. All three models returned numeric WER, CER and timing values for every clip, with no recorded baseline errors. No scores have been imputed. [N]

Model	WER	CER	Mean latency	Mean RTF	Switch WER	Zero WER
Sahara	7.44%	3.58%	2.95 s	0.5143	8.33%	40/64
MMS	37.91%	15.82%	3.58 s	0.3398	57.29%	14/64
Whisper	95.89%	45.54%	3.04 s	0.4195	91.15%	2/64

Switch-point scores use 32 annotated clips; the other baseline columns use 64. A zero-WER result agrees with the stored reference under the benchmark scoring convention.

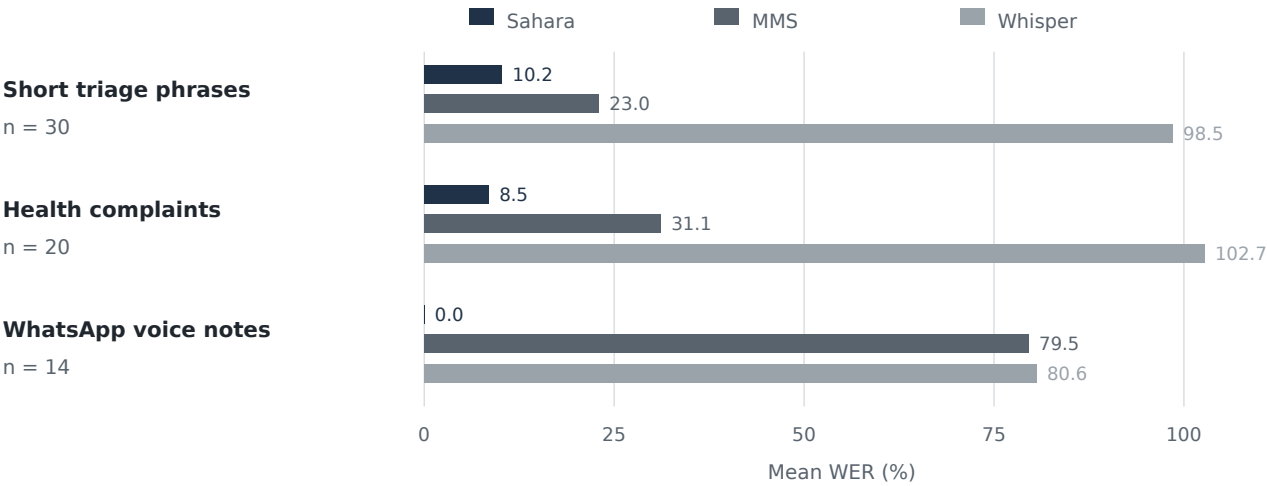


Figure 1. Mean baseline WER by recording group. All three models use the same clips within each group. Source: [N], by_recording_type.

Interpretation

Sahara is lowest in each recording group: **10.22%** on short phrases, **8.49%** on health complaints and **0.00%** on the 14 WhatsApp notes. The zero result means agreement with those stored references; it does not establish error-free transcription outside this sample.

MMS is the stronger local baseline, although its mean WER rises from **23.00%** on short phrases to **79.54%** on WhatsApp notes. Whisper records **98.50%**, **102.69%** and **80.58%** across the same groups. Device, speaker, channel and delivery style are not independently controlled.

At annotated language boundaries, Sahara records **8.33%** WER, MMS **57.29%** and Whisper **91.15%**. These boundary-local results reinforce the transcription comparison, but cover only the supplied annotations.

Tie-aware comparison

Across the baseline clips, Sahara has 44 outright WER wins and MMS has 2; 18 clips are tied for the lowest WER. Whisper has no outright wins. The single best_model field is not used to assign tied cases to a winner.

3. Controlled-noise recognition

The final sweep contains 64 scored clips for each model at each SNR. Higher SNR is cleaner; 5 dB is the strongest added-noise condition evaluated. Lines connect measured means only and are not estimates of untested conditions. [N]

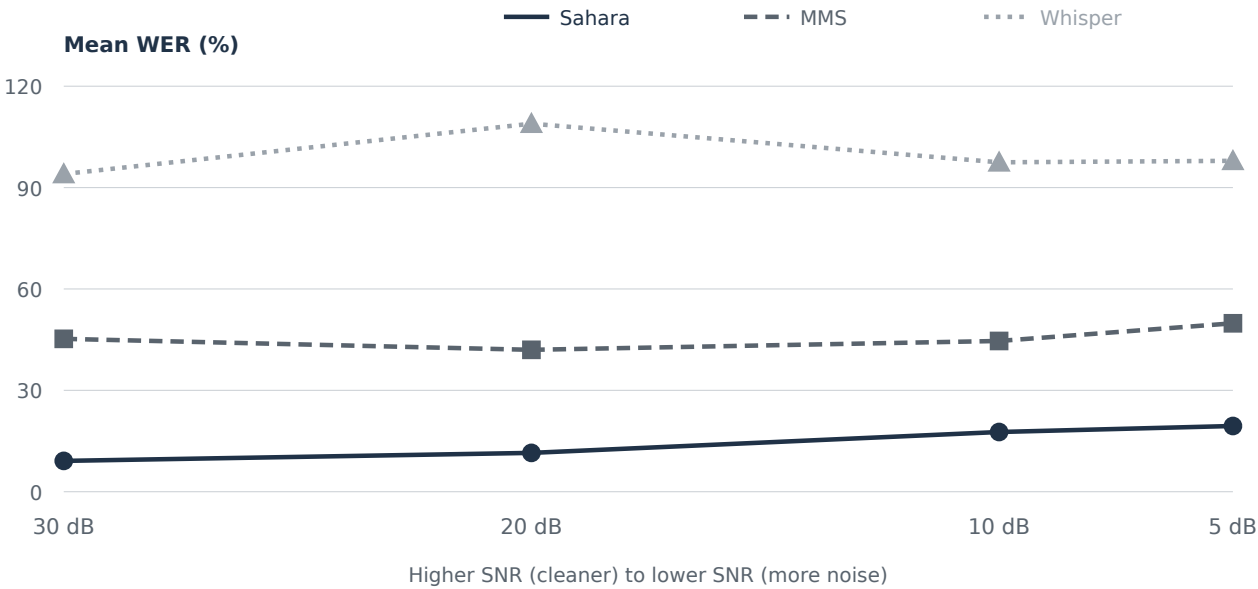


Figure 2. Mean WER across the four measured SNR levels. Marker shape and line style distinguish models without relying on color. Every point has n = 64.

SNR	Sahara WER	Sahara CER	MMS WER	MMS CER	Whisper WER	Whisper CER
30 dB	9.12%	4.92%	45.24%	17.64%	94.02%	45.60%
20 dB	11.49%	7.06%	41.99%	17.82%	108.91%	50.55%
10 dB	17.67%	10.70%	44.61%	19.62%	97.48%	56.65%
5 dB	19.45%	11.60%	49.84%	22.06%	97.92%	68.18%

What the completed sweep shows

Sahara: WER rises consistently from 9.12% at 30 dB to 19.45% at 5 dB, while CER rises from 4.92% to 11.60%. The unperturbed baseline is 7.44% WER. Sahara remains the lowest-error model at all four measured levels.

MMS: WER ranges from 41.99% to 49.84% and is lower than Whisper at every SNR. The WER trend is not strictly monotonic, although CER increases at each step as SNR falls. Its unperturbed baseline is 37.91% WER.

Whisper: WER remains high and non-monotonic, with its highest mean at 20 dB (108.91%). One three-word clip, `kelvin_sample_29.mp3`, records 866.67% WER at that level, illustrating how insertion-heavy short clips affect an unweighted mean. CER rises from 45.60% to 68.18% from 30 to 5 dB.

The same-export unperturbed Whisper baseline is 95.89% WER. A small change from an already high error rate should not be interpreted as reliable noise robustness. Full per-condition WER and CER are retained above.

4. Noise timing and run completeness

Model	Attempts	Valid results	Recorded errors
Sahara	256	256	0
MMS	256	256	0
Whisper	256	256	0

The final export contains numeric WER, CER and RTF for every clip-model-SNR record, with no recorded errors. This verifies recorded completion; it does not independently verify the injected noise characteristics. [N]

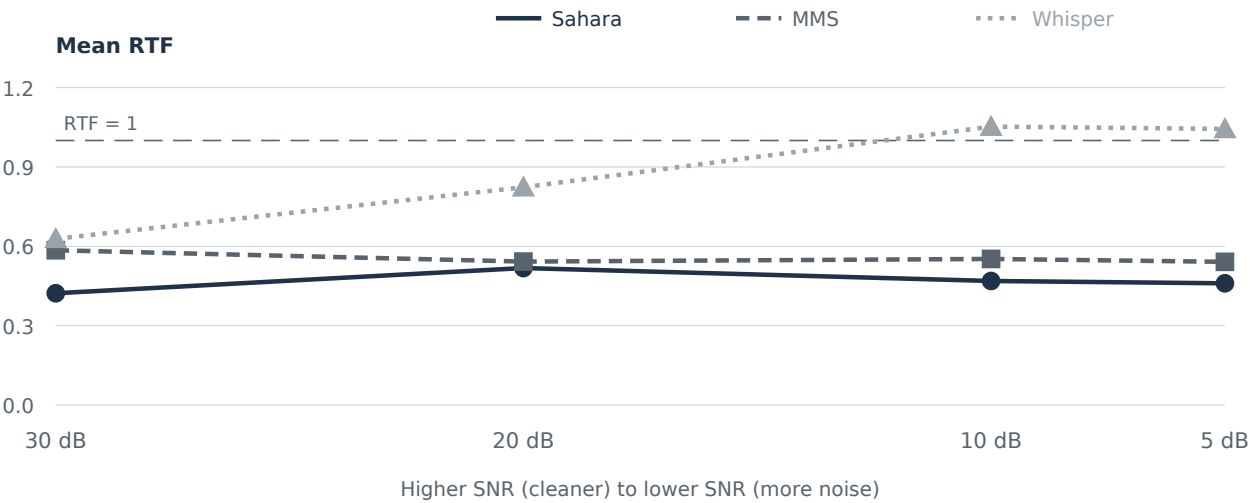


Figure 3. Mean real-time factor at each SNR. RTF = 1 is the audio-duration boundary. Timings belong to this machine and run, and the Sahara path includes remote-service interaction.

SNR	Sahara RTF	Sahara < 1	MMS RTF	MMS < 1	Whisper RTF	Whisper < 1
30 dB	0.4226	61/64	0.5852	53/64	0.6285	54/64
20 dB	0.5179	60/64	0.5423	55/64	0.8234	48/64
10 dB	0.4690	59/64	0.5523	54/64	1.0525	44/64
5 dB	0.4602	60/64	0.5414	54/64	1.0438	42/64

Sahara and MMS have mean RTF below 1 at all four levels, but some individual noisy clips exceed 1 for both models. Whisper exceeds mean RTF 1 at 10 and 5 dB. Average throughput and individual response time should therefore be interpreted separately.

Acoustic diagnostics and reproducibility

The recorded mean silence ratio is **55.73%** (median 45.35%; range 32.42%-87.37%). Mean pause count is **24.16**, and mean longest pause is **1.28 seconds**. These are exported diagnostics, not evidence of a particular noise source or mixing method.

The export does not specify the noise recording or generator, random seed, gain normalization, exact transform, or whether SNR is calculated over all samples or active speech. These details are needed to reproduce the acoustic experiment. No missing result was interpolated.

5. Local-only inference and run modes

The `offline_simulation` section of the core export contains complete local-only results for Whisper and MMS. All 128 records completed without an error. Sahara has no offline result and is not included in this comparison. [C]

Model	WER	CER	Mean latency	Mean RTF	RTF < 1	Zero WER
MMS	37.91%	15.82%	3.93 s	0.3847	64/64	14/64
Whisper	96.20%	42.11%	2.98 s	0.4440	57/64	2/64

MMS has 50 outright WER wins; Whisper has 7; the remaining 7 clips are tied. MMS also records RTF below 1 on all 64 clips, compared with 57 for Whisper. Tied clips are reported separately from outright wins.

The evidence establishes completion of the local inference paths. It does not record OS-level network isolation, model download/cache state, cold-start loading, or device specifications. Accordingly, **local-only simulation** is the supported description, rather than independently verified air-gapped execution.

Run interpretation

The clean, noise and local-only results are reported by evaluation condition. Scores and timings are not pooled across runs because execution paths and timing boundaries differ. The benchmark reports each measured result in the context in which it was obtained.

6. Data quality and final interpretation

Reference and normalization checks

One unresolved audio-reference pair. For kelvin_sample_10.mp3, the stored reference is *Najisikia vizuri zaidi sasa*, while all three latest baseline outputs are the English equivalent, *I am feeling much better now*. Each receives 150% WER. The JSON alone cannot show whether the audio, reference or output language is responsible. The score is retained unchanged. [N]

Word segmentation matters. For kelvin_sample_17.mp3, Whisper outputs *Ana kohoa sana*. against *Anakohoa sana*, producing 100% WER but 7.69% CER. Both metrics are needed to distinguish substantial recognition errors from token-boundary differences. [N]

This export has 64 unique filenames and no exact duplicate reference strings. The language mismatch above remains documented, and no reference transcript has been silently corrected or replaced.

Limits on generalization

Limit	Effect on interpretation
Collection coverage	The collection is tagged Swahili-English and Kenya, with 64 short recordings. Speaker counts and demographic strata are not provided. Generalization to other languages, accents or populations is not established.
Confounded conditions	Recording type, device, channel and delivery style vary together. Group differences cannot be attributed to one factor alone.
Reference independence	The manifests identify human ground truth, but annotation provenance and independent adjudication are not documented. Perfect agreement on a subset is not independent validation.
Statistical uncertainty	One finalized noise sweep is available. The report contains descriptive means, not confidence intervals, statistical significance claims or repeated-trial robustness estimates.
Configuration coverage	The exact Whisper checkpoint, decoding settings, full hardware and noise-generation recipe are absent from the exports. Findings apply to the recorded configurations.

Final conclusion

On this collection, Sahara provides the strongest recorded transcription performance and remains lowest-error across all four completed noise levels. MMS is the stronger evaluated local-only alternative. The complete noise sweep provides full coverage. Remaining limitations concern reference integrity and reproducibility, not missing noise measurements.

Appendix A. Methods and verification

This edition combines the complete noise run with the local-only evaluation. Each result is presented within its measured evaluation condition, and the calculations below preserve the stated denominators.

Source	File identity and timestamp	Fields used
Noise benchmark run	15 September 2026 13:24:40 UTC	manifest; overall; by_recording_type; by_clinical_category; clips; noise_robustness
Core benchmark run	15 September 2026 08:11:12 UTC	manifest; overall; clips; offline_simulation

Shared environment: schema 1.1-primary; Python 3.13.5; Windows-11-10.0.26200-SP0; collection data/primary_collection. Both manifests report commit 854d040f73e464ff903f6d05a059b0474f0c9eaf.

Verification performed for this edition

All baseline, core, noise and local-only summary WER, CER, RTF and latency values were checked against the stored per-result metrics where those aggregates are supplied. Differences are within the expected four-decimal rounding tolerance. Filenames and metadata match across the retained source runs.

Calculation	Rule
Completion	Count every clip-model-SNR record; require error = null and numeric WER, CER and RTF. Noise denominator: 64 per model per SNR; 768 overall.
Mean scores	Arithmetic mean of stored per-clip values. WER/CER are shown as percentages. No raw audio retranscription, independent edit-distance recomputation, interpolation or score replacement was performed.
Outright wins	Select the minimum per-clip WER across the evaluated models. Count an outright winner only when that minimum is unique; otherwise record a tie.
Switch-point WER	Use the export's stored boundary-window scores, with 32 numeric clip scores and 32 annotated windows. Window alignment is not independently reproduced.

Terminology: baseline = original, unperturbed recording, not necessarily noise-free audio; SNR = signal-to-noise ratio; RTF = real-time factor. Tables use the exported rounded means unless explicitly described as derived counts. All clinical-category names and benchmark labels are supplied metadata.

Appendix B. Clinical-category detail

Latest baseline mean WER by supplied category. These categories are descriptive metadata, not diagnoses or independently validated clinical strata. Aside from the phrase controls, each category contains only one to five clips; the table should not be used to rank performance across medical domains. [N]

Supplied category	n	Sahara WER	MMS WER	Whisper WER
Triage Phrase	30	10.22%	23.00%	98.50%
Respiratory	3	3.45%	52.02%	99.81%
Medication Side Effects	1	6.45%	35.48%	129.03%
Abdominal & GI	5	5.53%	71.65%	97.49%
Musculoskeletal	2	0.00%	53.46%	93.03%
Cardiac Emergency	1	12.90%	41.94%	93.55%
Paediatrics	3	0.93%	38.16%	63.22%
Dental & Oral	1	6.90%	37.93%	100.00%
Chronic Disease	4	3.25%	47.30%	83.96%
Respiratory Emergency	1	3.12%	34.38%	96.88%
Mental Health	2	3.84%	56.96%	95.57%
Eye & ENT	2	8.06%	59.01%	98.68%
Fever & Infection	2	3.33%	54.44%	79.45%
Trauma	2	15.88%	33.19%	101.72%
Urological	1	6.67%	33.33%	100.00%
Obstetrics	1	9.38%	25.00%	81.25%
Neurological	1	8.33%	8.33%	94.44%
Headache & Neurology	2	0.00%	93.75%	118.75%

Denominator check: category counts sum to 64. Each model is evaluated on the same clips within a category. WER above 100% is possible because insertions add to the edit count. Scores are reference-relative and do not measure clinical outcomes.

Reading small subsets

A 0.00% value in a one-clip category means one stored reference was matched. It is not evidence of complete coverage of that clinical domain. The aggregate transcription and noise sections provide the broader collection-level comparison.

Appendix C. Per-clip baseline WER

Short triage phrases

Exact filenames and latest unperturbed WER are retained for traceability. Values below are percentages, not accuracy scores. Source: [N], clips[].benchmark.results[].wer.

Filename	Sahara (%)	MMS (%)	Whisper (%)
kelvin_sample_1.mp3	0.00	0.00	0.00
kelvin_sample_2.mp3	0.00	0.00	50.00
kelvin_sample_3.mp3	0.00	25.00	100.00
kelvin_sample_4.mp3	0.00	50.00	0.00
kelvin_sample_5.mp3	0.00	66.67	133.33
kelvin_sample_6.mp3	0.00	0.00	166.67
kelvin_sample_7.mp3	0.00	0.00	100.00
kelvin_sample_8.mp3	0.00	0.00	75.00
kelvin_sample_9.mp3	0.00	33.33	83.33
kelvin_sample_10.mp3	150.00	150.00	150.00
kelvin_sample_11.mp3	0.00	0.00	100.00
kelvin_sample_12.mp3	0.00	20.00	100.00
kelvin_sample_13.mp3	20.00	20.00	100.00
kelvin_sample_14.mp3	0.00	0.00	100.00
kelvin_sample_15.mp3	0.00	20.00	80.00
kelvin_sample_16.mp3	0.00	40.00	100.00
kelvin_sample_17.mp3	0.00	0.00	100.00
kelvin_sample_18.mp3	0.00	25.00	100.00
kelvin_sample_19.mp3	50.00	33.33	100.00
kelvin_sample_20.mp3	0.00	0.00	100.00
kelvin_sample_21.mp3	0.00	20.00	80.00
kelvin_sample_22.mp3	0.00	0.00	250.00
kelvin_sample_23.mp3	20.00	0.00	120.00
kelvin_sample_24.mp3	0.00	25.00	125.00
kelvin_sample_25.mp3	0.00	0.00	120.00
kelvin_sample_26.mp3	0.00	0.00	80.00
kelvin_sample_27.mp3	0.00	75.00	75.00
kelvin_sample_28.mp3	0.00	20.00	100.00
kelvin_sample_29.mp3	0.00	0.00	100.00
kelvin_sample_30.mp3	66.67	66.67	66.67

Group mean (30 clips): Sahara 10.22%; MMS 23.00%; Whisper 98.50%.

The kelvin_sample_10.mp3 language mismatch is retained at 150% WER for all three models; see the data-quality section. This appendix does not change or reconstruct the ground-truth transcript.

Appendix C. Per-clip baseline WER

Health complaints

Exact filenames and latest unperturbed WER are retained for traceability. Values below are percentages, not accuracy scores. Source: [N], clips[].benchmark.results[].wer.

Filename	Sahara (%)	MMS (%)	Whisper (%)
sample_sw_en_Persisent_cough.m4a	3.45	37.93	131.03
sample_sw_en_Symptoms_after_taking_meds.m4a	6.45	35.48	129.03
sample_sw_en_adbominal_pain.m4a	17.65	38.24	94.12
sample_sw_en_backpain_explicit_negatives.m4a	0.00	35.48	96.77
sample_sw_en_chest_discomfort.m4a	12.90	41.94	93.55
sample_sw_en_child_described.m4a	2.78	13.89	83.33
sample_sw_en_dental_pain_swelling.m4a	6.90	37.93	100.00
sample_sw_en_diabetes_missed_food.m4a	3.33	46.67	133.33
sample_sw_en_diarrhoea_vomiting.m4a	10.00	36.67	110.00
sample_sw_en_difficulty_breathing.m4a	3.12	34.38	96.88
sample_sw_en_emotinal_distress.m4a	7.69	28.21	102.56
sample_sw_en_eye_exposure.m4a	16.13	22.58	106.45
sample_sw_en_fever_and_headache.m4a	6.67	20.00	103.33
sample_sw_en_head_injury_uncertainty.m4a	21.43	25.00	100.00
sample_sw_en_injury_ongoning_bleeding.m4a	10.34	41.38	103.45
sample_sw_en_medication-refill.m4a	9.68	29.03	83.87
sample_sw_en_mild_respiratory_symptoms.m4a	6.90	31.03	110.34
sample_sw_en_pain_urinating.m4a	6.67	33.33	100.00
sample_sw_en_pregnancy_bleeding.m4a	9.38	25.00	81.25
sample_sw_en_sudden_change_relative.m4a	8.33	8.33	94.44

Group mean (20 clips): Sahara 8.49%; MMS 31.13%; Whisper 102.69%.

The group name follows the supplied recording_type field. It does not establish that the recordings are unscripted, clinically adjudicated, or sampled from real patient encounters.

Appendix C. Per-clip baseline WER

WhatsApp voice notes

Exact filenames and latest unperturbed WER are retained for traceability. Values below are percentages, not accuracy scores. Source: [N], clips[].benchmark.results[].wer.

Filename	Sahara (%)	MMS (%)	Whisper (%)
WhatsApp Ptt 2026-09-14 at 8.33.26 PM.ogg	0.00	87.50	137.50
WhatsApp Ptt 2026-09-14 at 8.33.54 PM.ogg	0.00	83.33	83.33
WhatsApp Ptt 2026-09-14 at 8.34.13 PM.ogg	0.00	100.00	100.00
WhatsApp Ptt 2026-09-14 at 8.34.36 PM.ogg	0.00	95.45	90.91
WhatsApp Ptt 2026-09-14 at 8.34.48 PM.ogg	0.00	27.78	72.22
WhatsApp Ptt 2026-09-14 at 8.35.07 PM.ogg	0.00	100.00	100.00
WhatsApp Ptt 2026-09-14 at 8.35.23 PM.ogg	0.00	33.33	19.05
WhatsApp Ptt 2026-09-14 at 8.35.46 PM.ogg	0.00	85.71	88.57
WhatsApp Ptt 2026-09-14 at 8.36.04 PM.ogg	0.00	100.00	100.00
WhatsApp Ptt 2026-09-14 at 8.36.42 PM.ogg	0.00	88.89	55.56
WhatsApp Ptt 2026-09-14 at 8.37.02 PM.ogg	0.00	85.71	46.43
WhatsApp Ptt 2026-09-14 at 8.38.08 PM.ogg	0.00	67.27	87.27
WhatsApp Ptt 2026-09-14 at 8.39.00 PM.ogg	0.00	71.43	89.29
WhatsApp Ptt 2026-09-14 at 8.40.30 PM.ogg	0.00	87.10	58.06

Group mean (14 clips): Sahara 0.00%; MMS 79.54%; Whisper 80.58%.

Sahara matches all 14 stored WhatsApp references in this run. Speaker, codec, channel and background conditions remain confounded; this subset is not a general test of all WhatsApp audio.